Next-Generation Arm-based CPU FUJITSU-MONAKA for Green AI Data Center

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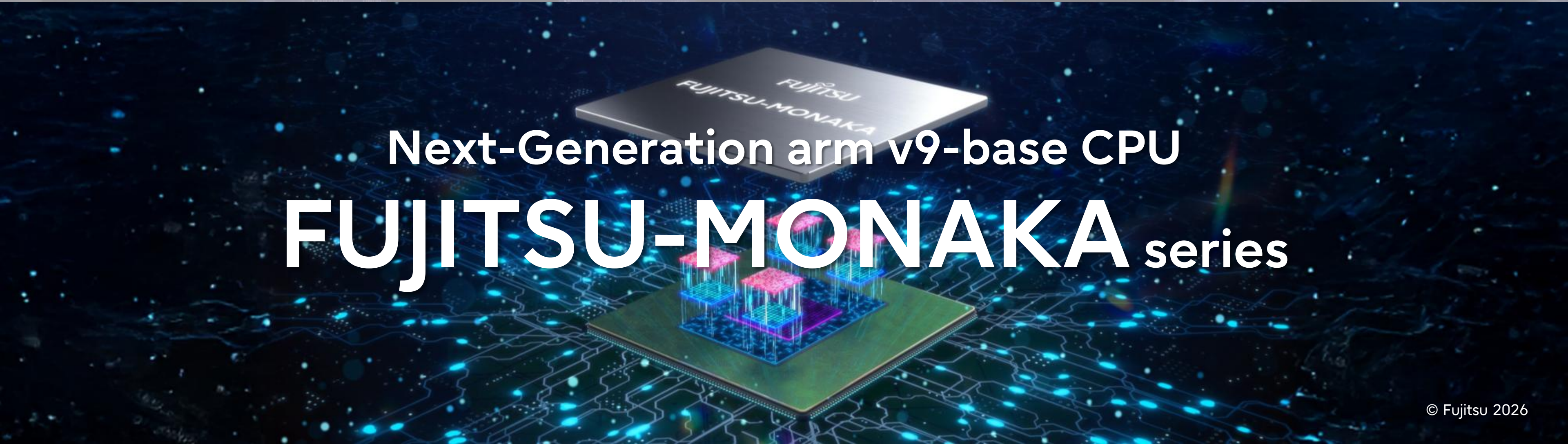
Technology Cultivated
Through the Development of World-Class Supercomputers

Next-Generation arm v9-base CPU

FUJITSU-MONAKA series



CPUs as the Foundation for Evolving AI/HPC Systems

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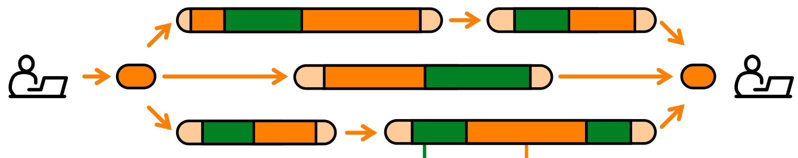
In AI/HPC systems, there are roles that only the CPU can fulfill

- AI: Agentic AI drives greater CPU usage for orchestration and peripheral processing
  CPU processing is also required for general workloads driven by agentic AI
- HPC: CPUs remain essential for high-throughput and high-precision workloads

AI Workloads

Multi-Agent collaborative execution



AI workload CPU GPU

LLM, Speech, Video, Vision, etc.

General-purpose workloads CPU CPU

Orchestration, RAG, DB, Code execution, etc.

HPC Workloads

Precision Small-Job Processing

CPU – Small jobs x many

Startup ≈ compute → CPU

startup compute

CPU – High precision

FP64

3.14159265

Large-Scale Parallel Processing

GPU – Large jobs x few

Startup << compute → GPU

compute

startup compute

GPU – Speed over precision

FP16/FP32

3.1415

FUJITSU-MONAKA Key Message and Value Proposition

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Key Message

- AI's increasing widespread adoption is constrained by power and cost.
  FUJITSU-MONAKA serves as a solution to overcome these constraints.

Key Trends

Workload diversification evolving compute demands

Power constraints

Sovereign requirement

FUJITSU-MONAKA Value

Up to 2× AI Performance powered by architecture and an open ecosystem

Over 50% TCO reduction by ultra-low-voltage

Secure In-Use Data Protection with Confidential Computing

FUJITSU-MONAKA Processor Overview

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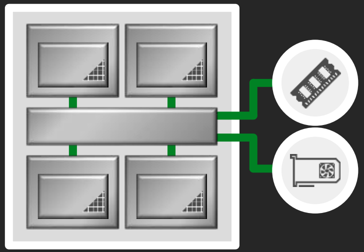
FUJITSU-MONAKA



- Arm® v9.3-A Architecture
- 3D chiplet
  - Core die 2nm
  - SRAM die/IO die 5nm
- Ultra low voltage for energy-efficiency
- DDR5 12 channels 8000+MT/s
- Liquid / Air-cooling
- Arm SVE2-256bit for AI and HPC
- 144 cores x 2 sockets (288 cores per node)
- Confidential Compute Architecture for security
- PCI Express 6.0 (CXL3.0) 96 lanes

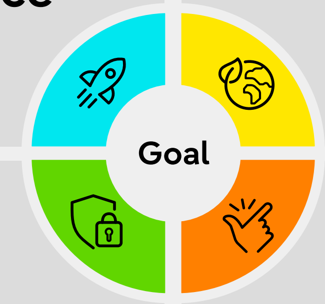
subject to change without notice

**Evaluation samples start shipping from NOW.
Product shipments begin in 2027.**

A Made-in-Japan CPU Engineered for AI Performance and Power Efficiency

High-Performance

Power-Efficiency



Safety & Security

Ease of Use

Semiconductor Trends -High Performance-

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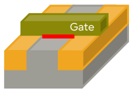
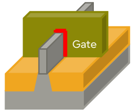
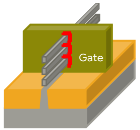
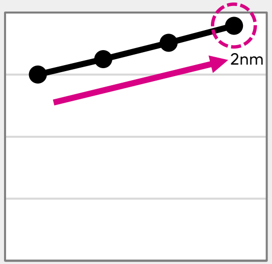
- **(Pros.)**
  Continuing improvements in semiconductor performance and power consumption, due to advancements in transistor structures
- **(Cons.)**
  Facing challenges with the slowdown in increasing cache memory capacity and the drastic increase in cost

	Transistor Type		
	Planar	FinFET	GAA (Gate-All-Around)
Technology node	~20nm (K computer：45nm)	16nm ~ 3nm (Fugaku：7nm)	2nm~ (MONAKA：2nm)
Semiconductors Structures			

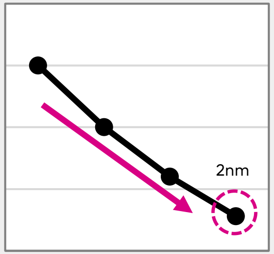
Semiconductor Trends*

*Fujitsu estimation

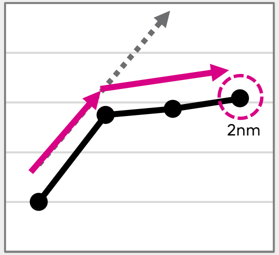
Frequency



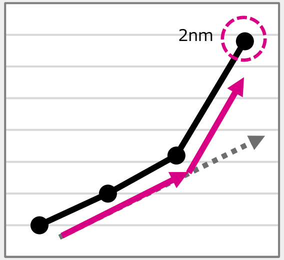
Power Consumption



Cache Memory Capacity



Cost



2nm monolithic design is NOT the optimal solution to meet all the requirements for power, performance, and cost.
We have adopted an innovative chiplet-based architecture for the FUJITSU-MONAKA project, allowing us to select the most suitable technology for each design component.

3D Microarchitecture to Overcome Cost Constraints

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- **FUJITSU-MONAKA adopts the innovative 3D many-core architecture**

  Core die
  - TSMC-N2P
  - Cores in 2nm Core die to achieve high performance and low power

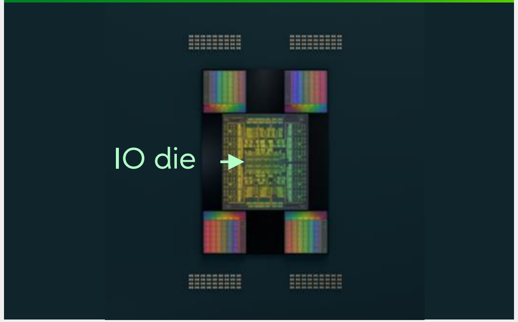
  SRAM die
  - TSMC-N5
  - All the last level cache are in 5nm SRAM die (bottom die), tightly coupled with core die through Hybrid Bonding

  IO die
  - TSMC-N5
  - All IOs are in stable 5nm IO die, connected with SRAM die via Silicon Interposer

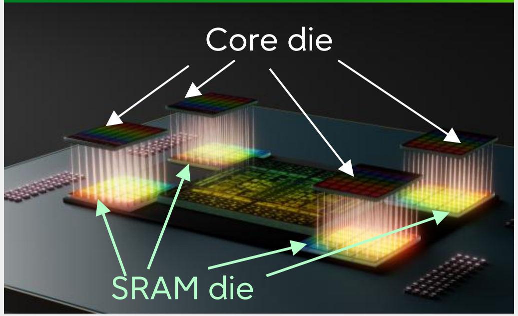
2nm silicon area is < 30% in total silicon area, which contributes to manufacturing cost reduction and better performance balance.

By minimizing 2nm macros and placing them in the stable 5nm SRAM die as much as possible, we accelerated time-to-market for the 2nm chip.

Top View



Perspective view



3D Integration: Cooling, Latency and Power Delivery

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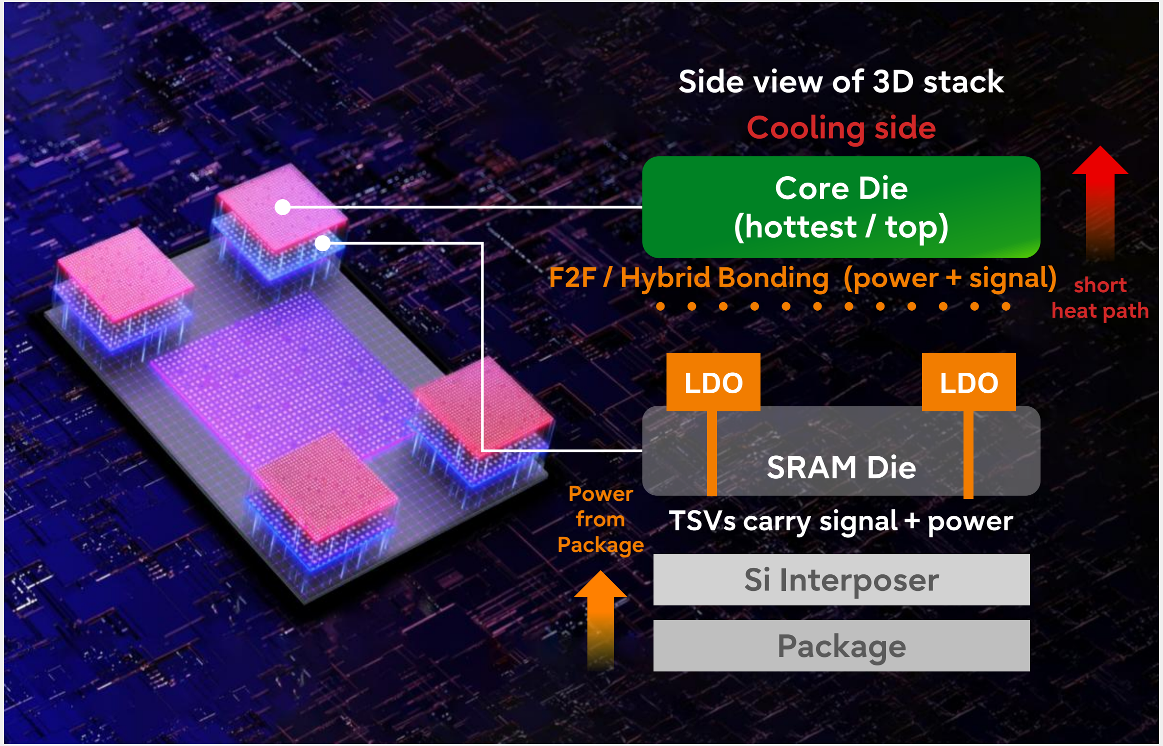
Top-side cooling + direct F2F/Hybrid Bonding + local LDOs solve thermal, latency and DVFS constraints



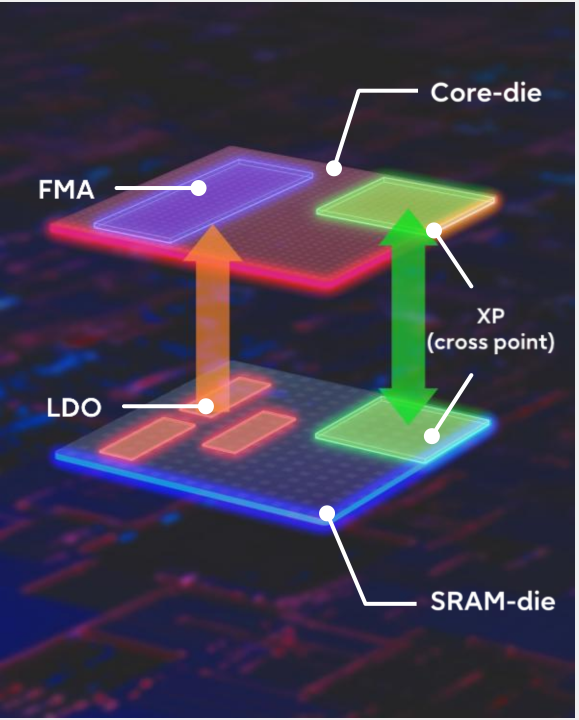
1. **Core Die on cooling side**
   Core Die is hottest; top placement shortens the thermal path to the Cooling side

2. **F2F/Hybrid Bonding reduces latency**
   Direct F2F/Hybrid Bonding connection shortens the Core Die–SRAM Die signal path

3. **LDO enables per-core DVFS**
   Local LDOs provide regulated supply for per-core voltage and frequency control

4. **3D Power delivery route**
   Package → Si Interposer → SRAM Die(TSV → LDO) → Hybrid Bonding → Core Die

LDO Placement Optimization of 3D Micro Architecture



- Core die is a repeating succession of cores, and SRAM-die also has a repeating structure of Last Level Cache, connected via XP (cross point)
- **Benefit 1:**
  LDOs are analog circuits and, similar to SRAM, have poor shrink rates. By placing such circuits on an SRAM die that uses 5nm technology, we achieve optimal cost-performance.
- **Benefit 2:**
  By strategically placing LDOs beneath high-power consumption circuits, such as an FMA (Floating Multiple-Add) unit, power can be supplied more efficiently and stably.

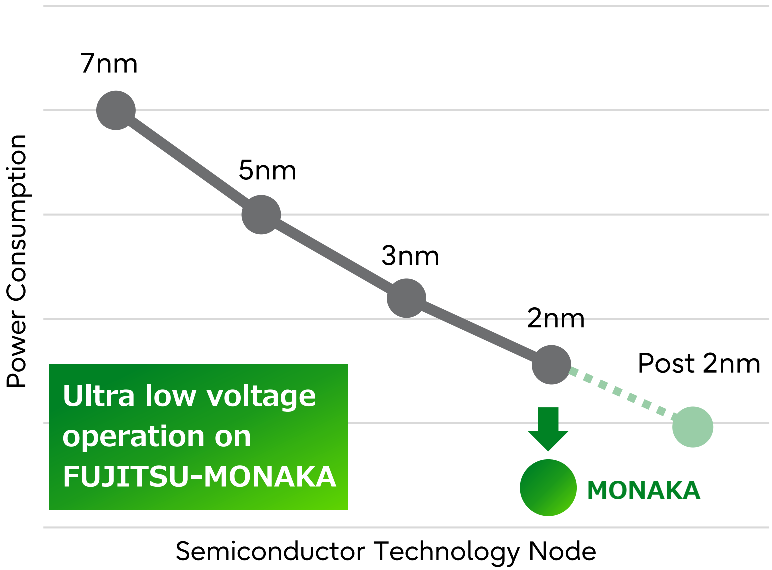
Ultra Low Voltage Technology to Overcome Power Constraints

subject to change

- FUJITSU-MONAKA's ultra low voltage operation technology enables **energy saving comparable to one generation ahead of 2nm**

Trend of Semiconductor Power*

*Fujitsu estimation



- Reducing power consumption by lowering voltage of the CPU30% lower than nominal voltage = 50% lower power consumption

$$P \propto C\, V^2 f$$

C : Capacity
V : Voltage
f : Clock Frequency

- Fujitsu develops **custom circuits including SRAM by using our proprietary CAD**, which enables stable operation at an ultra low voltage

*Fujitsu estimation

Fujitsu's Proprietary Core Design

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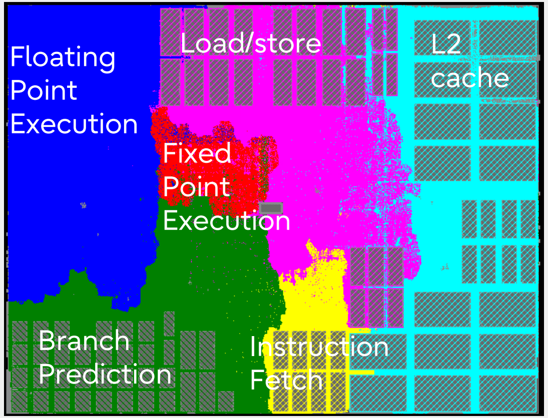
- **Arm v9.3-A compliance FUJITSU microarchitecture**
  - PPA optimization in microarchitecture level for AI/HPC workload
- **A core microarchitecture optimized for low power consumption**
  - Development of a custom evaluation CAD tool at Fujitsu for non-standard ultra-low voltage operations and tuning of pipeline stages to meet stringent frequency requirements
  - Stable operation at ultra-low voltage using custom-designed SRAM
- **Enhanced SVE2 engine**
  - Two 256-bit width SVE2 execution units
  - Supporting additional data type, FP8/INT8MM, for AI inference workload
  - Two 256-bit load/store units to align SVE2 width
- **For general purpose workload**
  - Three-level advanced TAGE-based branch prediction
  - Six ALUs to improve general-purpose workload processing performance
- **Mainframe-level RAS for low CPU failure rate**
  - ECC or duplication protection in L1/L2 cache
  - parity check in execution units and registers
  - Hardware instruction retry mechanism to recover from transient error



~1.47 mm2

Fujitsu's Proprietary Core Pipeline Overview

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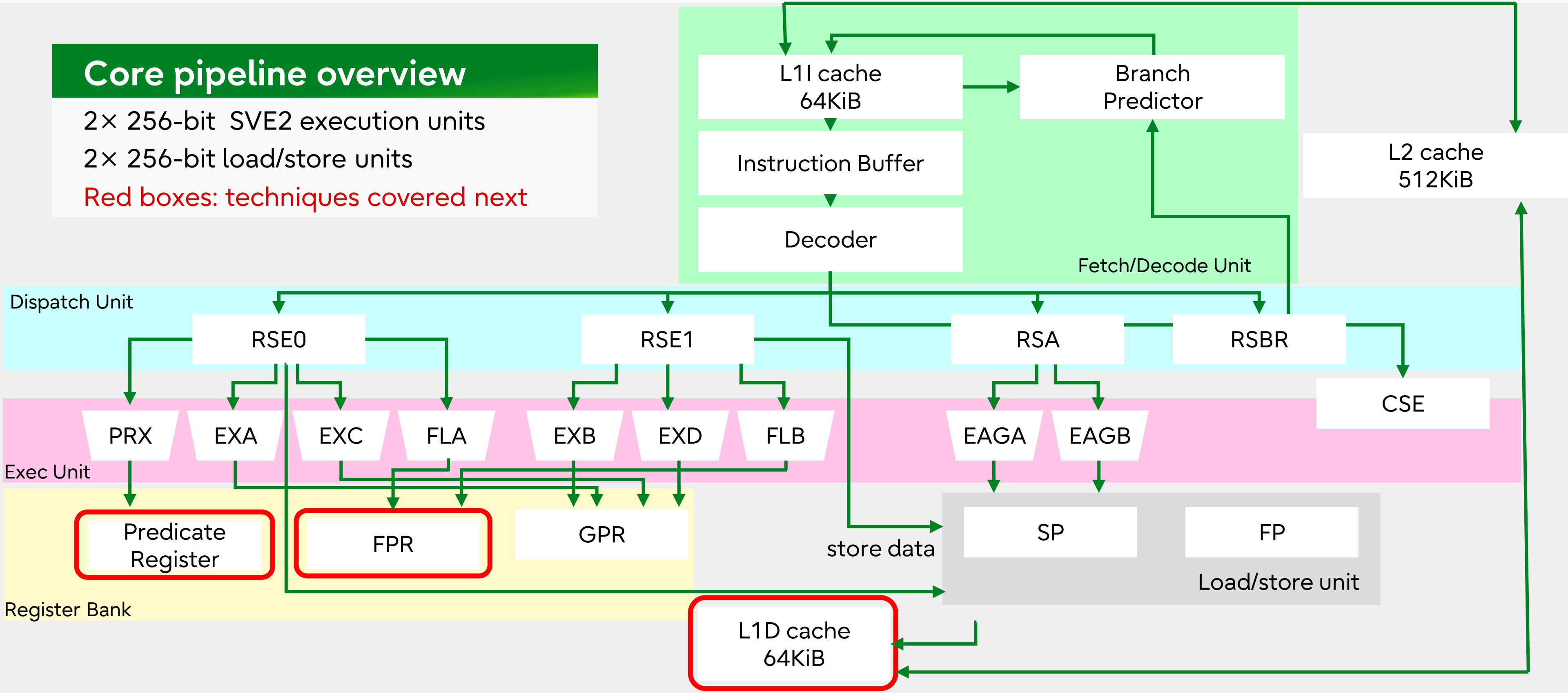
Core pipeline overview

2× 256-bit SVE2 execution units

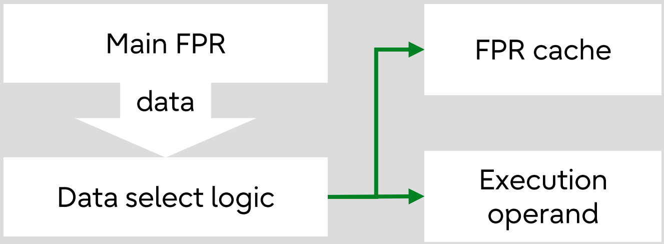
2× 256-bit load/store units

Red boxes: techniques covered next

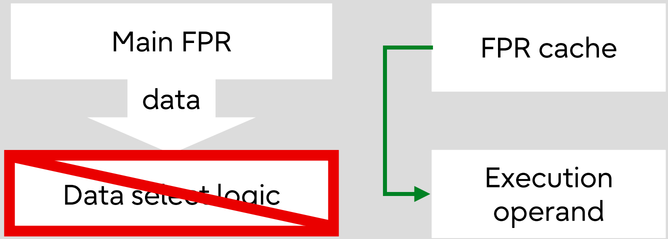


Exploring Power Optimization Techniques in Core Microarchitecture(1/3)

- FPR(Floating Point Register) cache.
  Programs with high temporal locality, such as GEMM, benefit significantly from power reduction.



Read:
The FPR cache temporarily holds FPR data.



Read again:
When the data is available in the FPR cache, access to the Main FPR can be avoided, reducing power consumption.

Exploring Power Optimization Techniques in Core Microarchitecture(2/3)

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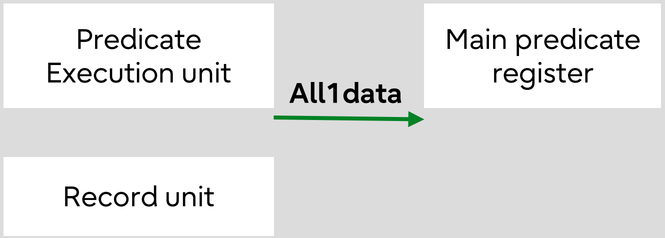
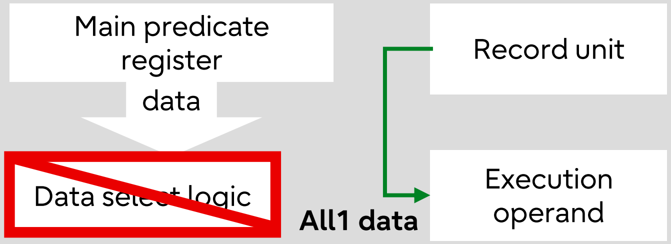
- Reduction of read power consumption in Predicate register.
The predicate data pattern is often active (all 1’b), which makes this technique widely applicable.



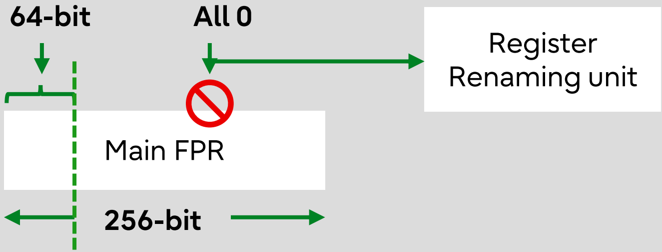
Write:
When writing common data patterns (e.g., All 1'b) of a predicate to a predicate register, store that information in a record unit as well.



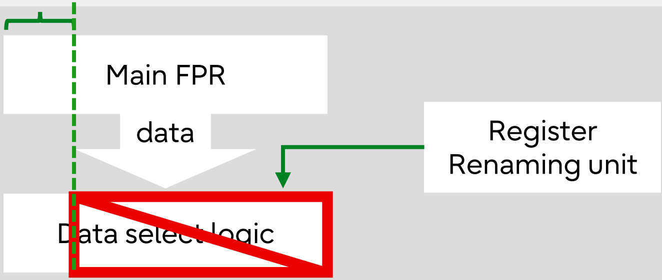
Read:
Common predicate patterns (e.g., All 1’b) are supplied from the Record Unit. Accesses to the Main Predicate Register are avoided, reducing read power consumption.

Exploring Power Optimization Techniques in Core Microarchitecture(3/3)

- Reduction of power consumption when data size is less than FPR length.
  This is effective when running Arm NEON code.



Write:
Many NEON operations use only 64-bit or 128-bit data in a 256-bit FPR. The Register Remaining unit records the unused upper bits are set 1’b0, eliminating unnecessary zero writes and reducing write power.



Read:
The Register Renaming unit provides the zero information for the unused upper bits. Accesses to the upper-bit data path are skipped, reducing read power.

Exploring High Throughput Techniques in Core Microarchitecture(1/1)

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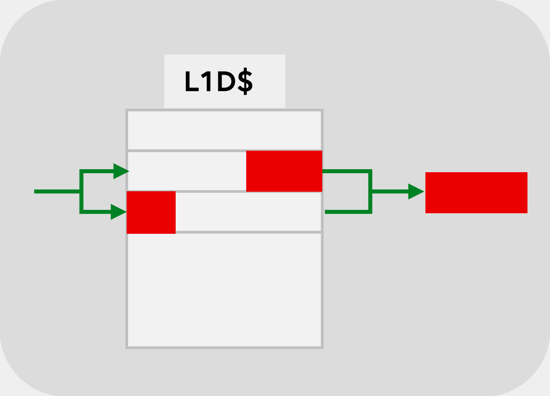
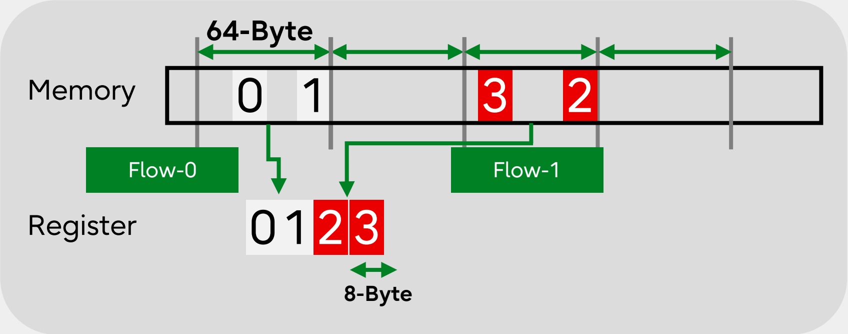
- To effectively utilize the 256-bit SIMD unit, leveraging HPC technical expertise to maximize L1 cache throughput.

1. Sustained throughput for 256-bit SIMD load
   - An unaligned SIMD load crossing cache line keeps the same throughput



2. " Combined Gather" mechanism increasing gather throughput
   - Gather processing is important for real HPC applications
   - FUJITSU-MONAKA continues to use "Combined Gather" mechanism enabling to return up to two consecutive elements in a "64-byte aligned block" simultaneously.
   - This mechanism doubles the throughput of the core's gather instruction.



Memory Access - NUMA Node in FUJITSU-MONAKA

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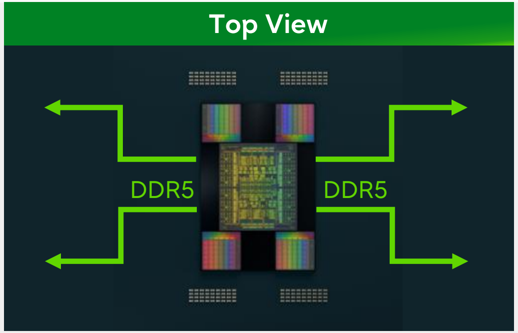
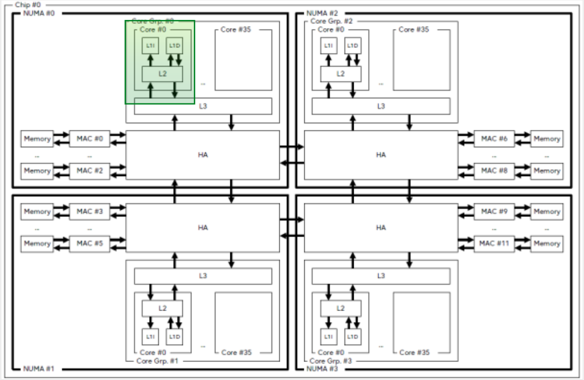


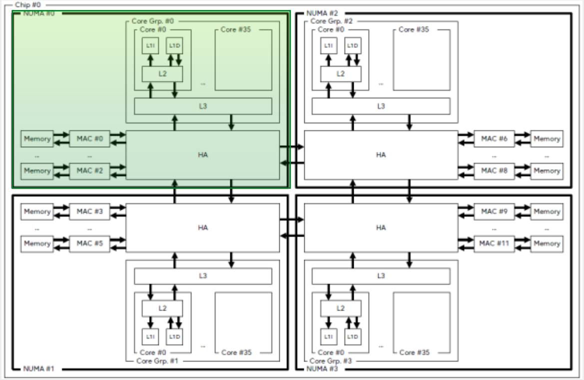
NUMA node can be flexibly configured to align with the desired software characteristics.

- 18 core x 8 NUMA node : Last Level Cache Latency/Throughput
- 36 core x 4 NUMA node : Memory Latency/Throughput balancing
- 144 core x 1 NUMA node : Large size memory application

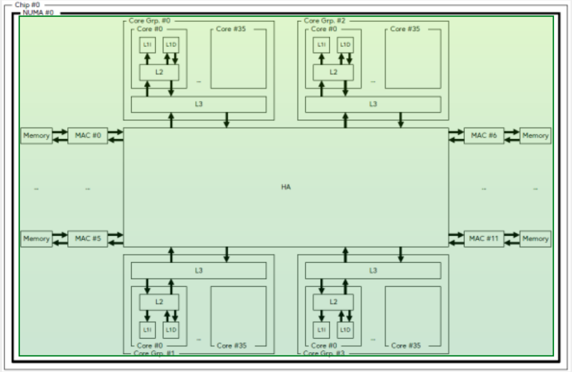
18core x 8 NUMA node



36core x 4 NUMA node



144core x 1 NUMA node



: NUMA node

FUJITSU-MONAKA Memory Hierarchy

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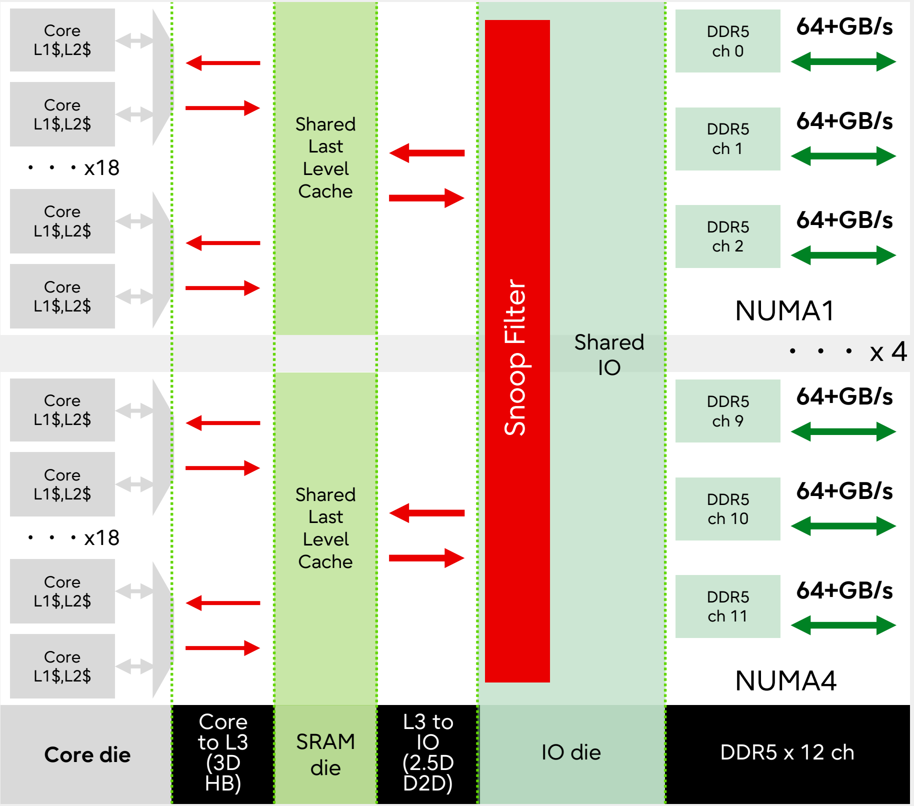

3 Key Points in FUJITSU-MONAKA

1. **Memory access localization**
   - Allows up to 4 NUMA regions
   - Each NUMA node has the nearest L3 & 3xDDR Ch
   - Allows dividing two L3 regions within each NUMA mode(SRAM-Die Divided Mode)
2. **Memory partitioning for QoS regulation**
   - Up to 63 MPAM ID(FEAT_MPAMv1p1)
   - MPAM configuration is independent of NUMA nodes
3. **Data granularity alignment "from Core to End"**
   - Core pipeline : 32byte(256bit) SVE2 x 2 pipelines
   - Cache Memory : 64byte L1/L2/L3 cache line size
   - Data traffic : 32byte x 2 data fabric
   - Memory endpoint: DDR5 64byte burst size

Confidential Computing in FUJITSU-MONAKA

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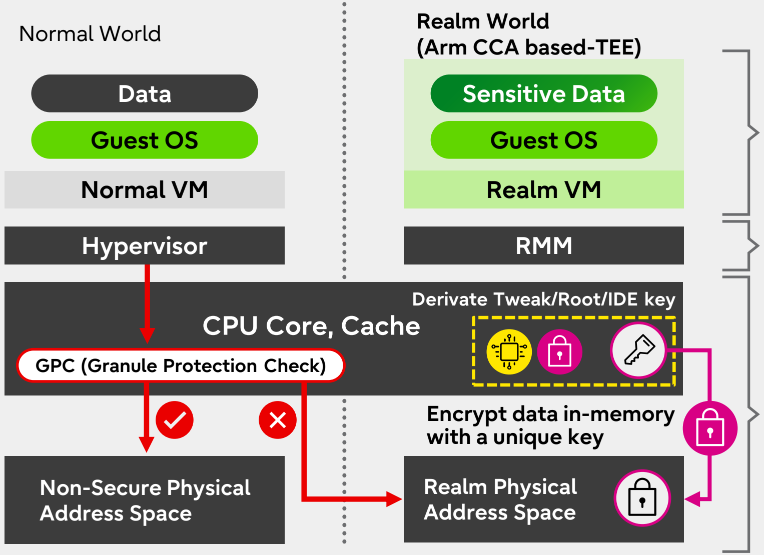
Fujitsu delivers a secure platform to meet Sovereignty requirements

- **Security**: Protects data-in-use within VMs, using Arm Confidential Compute Architecture (CCA)
- **Integrity**: Prevents environment spoofing through hardware-based remote attestation
- **Compatibility**: VM-based CCA technology enables seamless migration of existing environment



Sovereignty: Fujitsu delivers CCA Full-Stack platform

Software Stack

Confidential VM / Container Guest Kernel, CoCo
Remote Attestation Veraison, Trustee
Host OS/ Hypervisor Host Kernel, QEMU, EDK2, libvirt, OpenStack, KubeVirt

Firmware

TF-A: Establishes the trusted boot chain for RMM and Secure Monitor
TF-RMM: Manages Realm lifecycle, memory, and attestation

Hardware Root of Trust (FUJITSU-MONAKA)

Security ensured by:
- Sub processor HW

Tweak/Root/IDE key
Derivation core/cache

Encrypt data
for out of chip

FUJITSU-MONAKA Software Ecosystem

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Seamless Arm v9 ecosystem adaption with MONAKA and NVIDIA HPC SDK

- Fujitsu actively contributes to the development of the Arm v9 OSS ecosystem.
- Expand AI & HPC acceleration frameworks and libraries with NVIDIA

Legend: OSS / ISV or OSS support / NVIDIA

	AI	DC	HPC	NVIDIA
FW・Library	AI FW: TensorFlow, ONNX, PyTorch	Data Analysis: Redis/Valkey, Apache Spark, PostgreSQL, TiDB, Clickhouse	Parallel exec. Environment: ucx, Open MPI, ucc with xCCL	HPC-X; HPC SDK: NCCL, Open MPI nvCC, nvFORTRAN CUDA, cuBLAS, …; Nsight, cuda-gdb
	Toolchain: LLVM, GCC, UXL (OneDNN, OneDAL,…), ArmPL, OpenSSL, Arm Compute Library, Arm Optimized Routines, OneTBB, OpenMP, OpenBLAS, FFTW3			
Service・App dev.	Inference serving: ollama, vLLM, Triton; RAG; ML/LLMOps: MinIO, MLflow, LangChain	CCA: CoCo, Trustee, Veraison; Monitoring: Grafana/Loki; Web: Nginx, Nginx plus; Service Development: Quetz, pypiserver, Jupyter Notebook	Perf. Anal. tools: perf, Linaro Forge; Job/system operation management: Slurm, AltairPBS pro, Ansible, Singularity, foreman, ISV cluster mng. software	
Operation management	Container Infra.: Kubernetes, Harbor, Kubernetes ISV distro.	Virtual Infra.: QEMU/KVM, OpenStack, OpenStack ISV distro., KubeVirt		Container Toolkit
OS・file system	Object Storage: Ceph, Ceph ISV distro.	Distributed filesystem: HDFS, Lustre, DDN EXAScaler		
	Distribution: RHEL, SLES, Ubuntu, Rocky Linux			DCGM, GPU Driver

Frequency / TDP / Benchmark

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	High efficiency SKU	High performance SKU
Number of core per socket	144	144
TDP	350W	500W
cooling	Air cooling	Liquid cooling
Base frequency	2.1GHz	2.9GHz
DGEMM(estimated)	4355 GFLOPS (@2.1GHz)	6013 GFLOPS (@2.9GHz)
STREAM Triad(estimated)	500 GB/s (@8.0GT/s)	
INT8(estimated)	69.7 TOPS(@2.1GHz)	96.2 TOPS (@2.9GHz)

- Thanks to ultra-low-voltage technology, even demanding programs like GEMM can operate at 2.1GHz across 144 cores while staying within 350W
- Per-core DVFS automatically optimizes power, performance, and frequency
- FUJTISU-MONAKA can operate at a high frequency , supporting programs that prioritize single-thread performance

"FUJITSU-MONAKA" Power-Efficient High-Performance CPU Development Journey

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Development Roadmap

2020

HPC, AI

7nm

- Adopted in the supercomputer Fugaku
- First in the world to implement Arm SVE
- Low voltage operation

2027

Data Center, HPC, AI

2nm

- Arm SVE2 support
- 3D many core
- Ultra-low voltage operation
- CCA

2029

Data Center, HPC, AI

1.4nm

- SME2 for AI
- Selected for FugakuNEXT
- NVLink™ Fusion for AI/HPC

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Thank you

Abbreviations

- RSA: Reservation station for address generation
- RSE: Reservation station for execution
- RSBR: Reservation station for branch
- GPR: General-purpose register
- FPR: Floating-point register
- CSE: Commit stack entry
- EAG: Effective address generator
- EX: Integer execution unit
- FL: Floating-point execution unit
- PRX: Predicate execution unit
- SP: Store port
- FP: Fetch port